

Sustainable Fishery Management in Odisha



Background of the Study

- Fisheries support livelihoods and food security for many coastal communities in Odisha.
- Increasing fishing pressure and environmental changes are stressing marine resources.
- Sustainable fisheries management is essential to protect fish stocks and ensure long-term livelihood security.

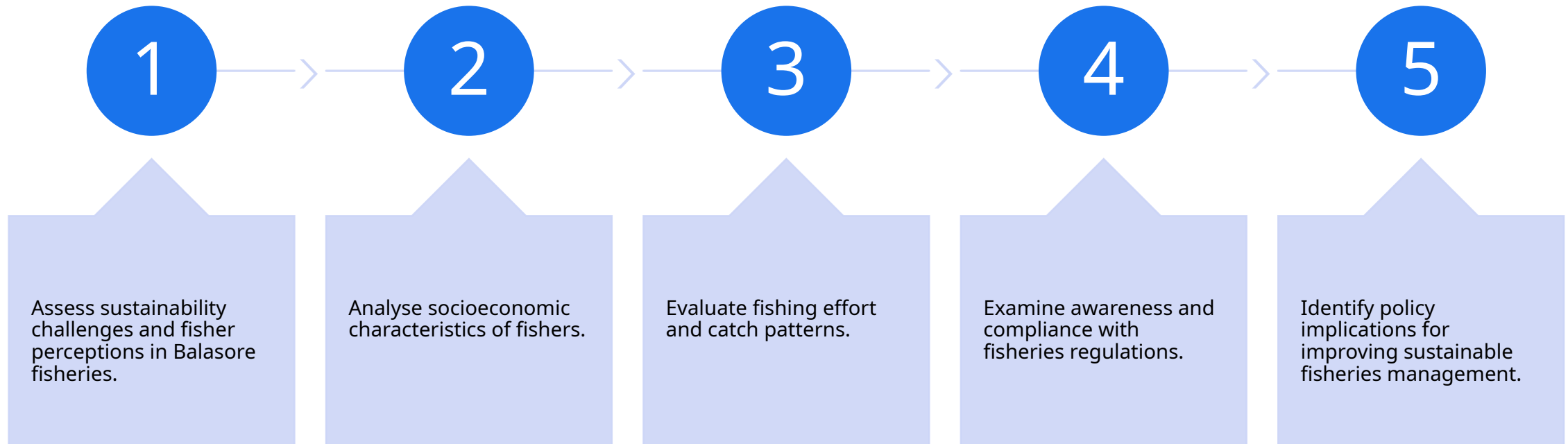




Research Problem

- Coastal fishers report declining fish stocks, raising sustainability concerns.
- Fishing households in Balasore face economic vulnerability due to fluctuating catches and unstable income.
- Limited regional research exists.
- Understanding fisher perceptions, practices, and regulatory compliance is essential for effective fisheries management.

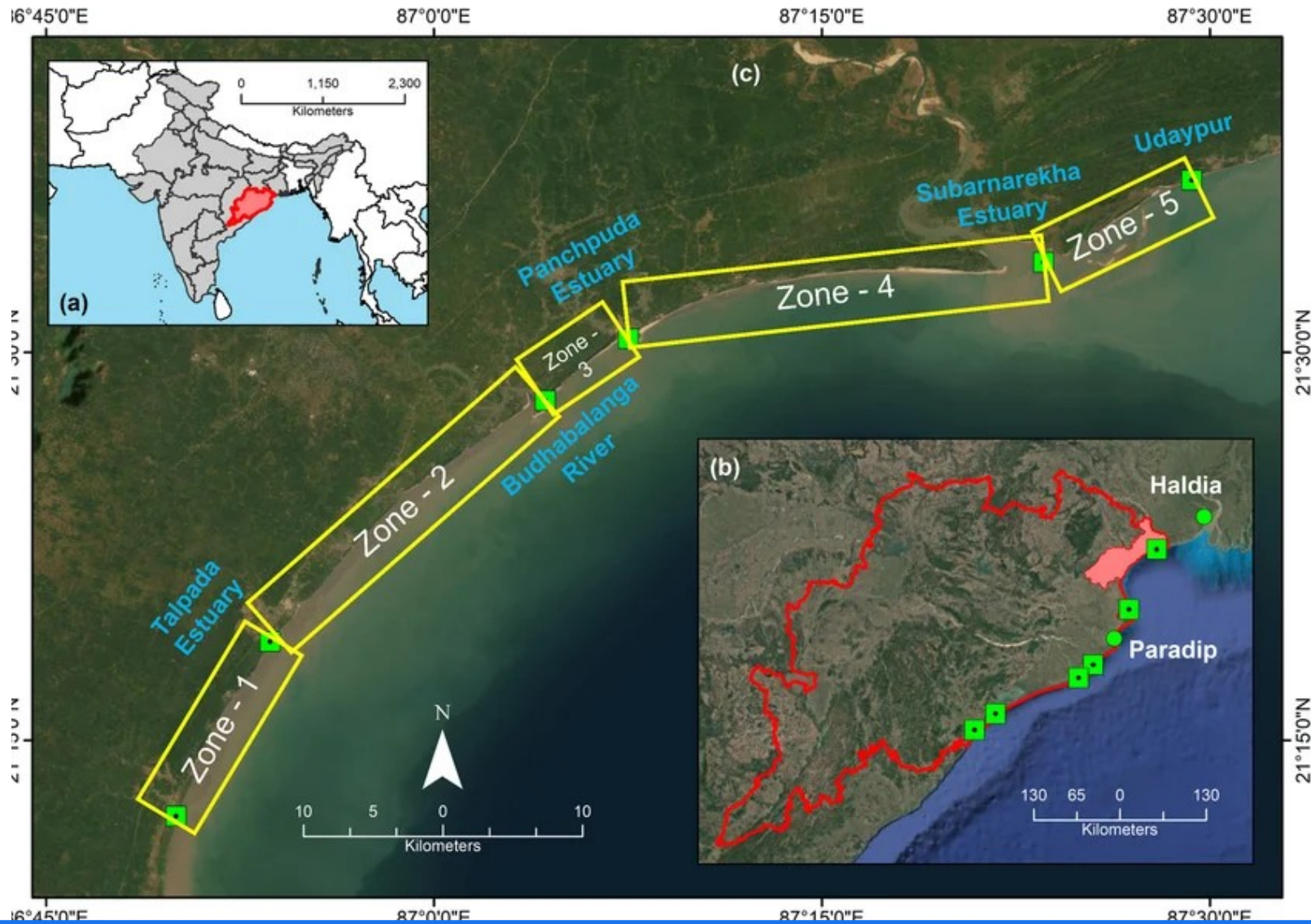
Research Objectives



Research Questions

This study addresses four key research questions:

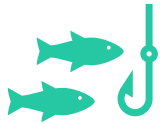
- What are the socioeconomic characteristics of fishers in Balasore?
- How dependent are fishing households on fisheries income for their livelihoods?
- To what extent are fishers aware of and compliant with sustainability regulations?
- What changes do fishers perceive in fish availability and overall marine resource conditions over time?



Study Area

- Balasore district in Odisha lies along the Bay of Bengal and has important traditional fishing communities.
- Marine fisheries support local livelihoods through employment, income generation, and food security.
- The study focuses on selected villages where households depend heavily on marine fishing activities.

Methodology Overview



The study adopts a quantitative analytical approach using primary survey data collected from 70 fishing households in Balasore district



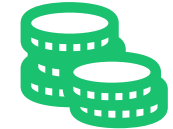
A production function framework is used to examine the relationship between fishing inputs and output (catch)



The analysis captures how effort, capital, and socioeconomic factors influence fish production



Data analysis includes descriptive statistics, correlation analysis, and functional relationship assessment



The approach enables linking economic behavior of fishers with sustainability outcomes

Analytical Framework

1

Socioeconomic factors such as age, education, and experience influence fishing effort and activity levels.

2

Fishing effort affects catch quantity and income generation.

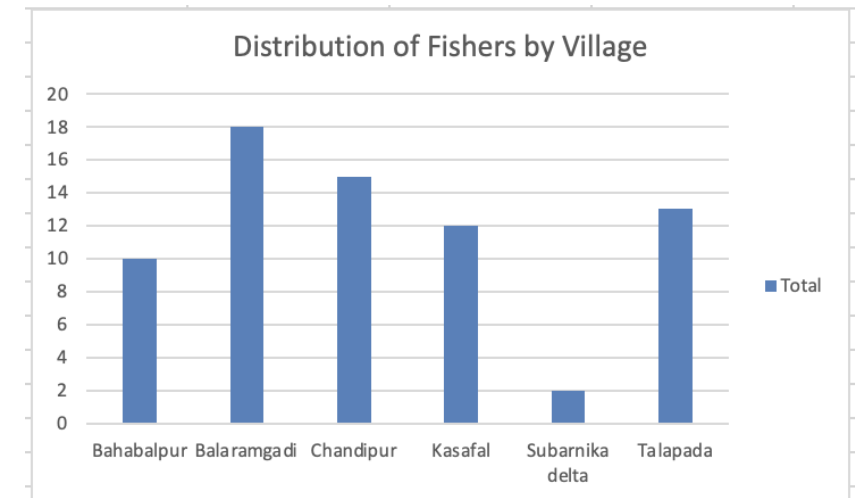
3

These outcomes shape perceptions of resource sustainability and inform fisheries management policies.

Distribution of Respondents by Village

- Bar chart shows distribution of surveyed fishers across villages in Balasore district.
- Most respondents come from coastal villages.
- Indicates strong geographical dependence of communities on marine fisheries for livelihood and economic activities.

Row Labels	Count of Q1_Village
Bahabalpur	10
Balaramgadi	18
Chandipur	15
Kasafal	12
Subarnika delta	2
Talapada	13
Grand Total	70



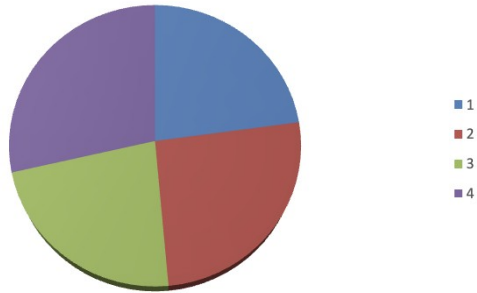
Socioeconomic Profile of Fishers and Fishing Experience and Effort

- Most fishers belong to the active working-age group, indicating strong dependence on fishing for livelihood.
- Education levels are generally low, which may limit awareness of fisheries regulations, sustainable practices, and alternative livelihood opportunities.
- Most respondents have many years of fishing experience, showing strong dependence on fisheries as a traditional livelihood.
- Fishing trips and fishing days vary among households, reflecting differences in fishing capacity, resource access, seasonal conditions, and economic needs.

Column Labels					
	20-30	31-40	41-50	51-60	Grand Total
Count of Age_Group	12	12	25	21	70

Row Labels	Count of Q4_Education
1	16
2	18
3	16
4	20
Grand Total	70

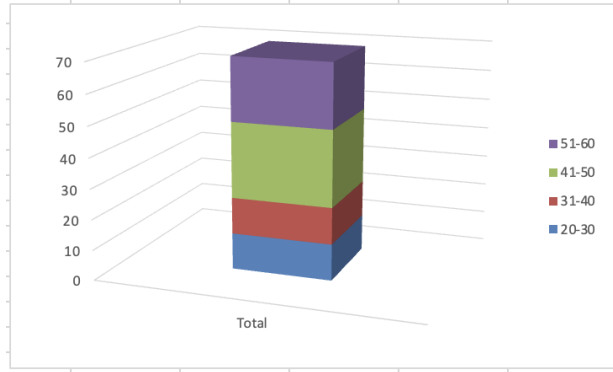
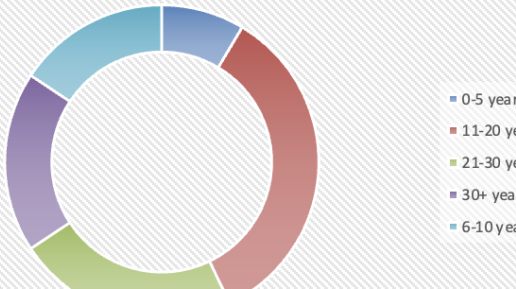
Education Level of Fishers



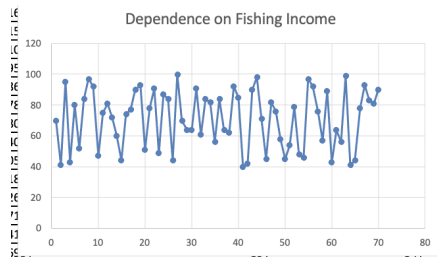
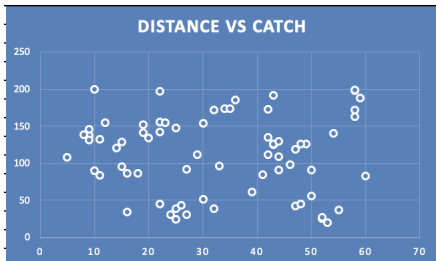
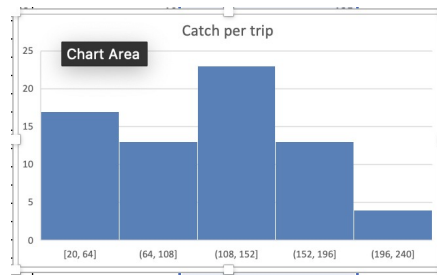
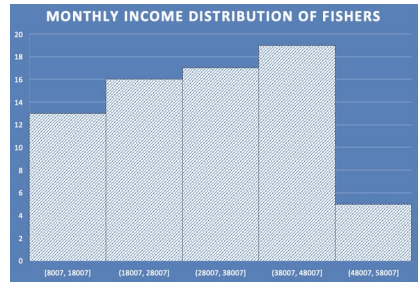
Average Catch per Trip by Village



Experience in fishing



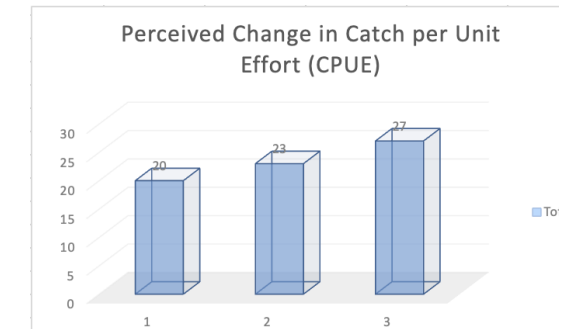
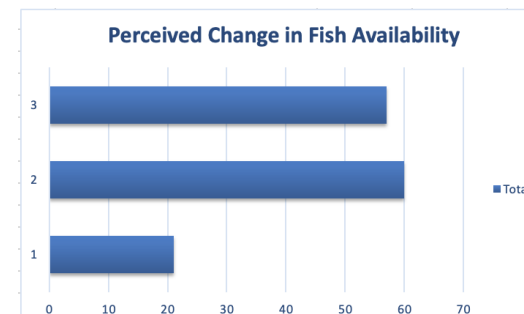
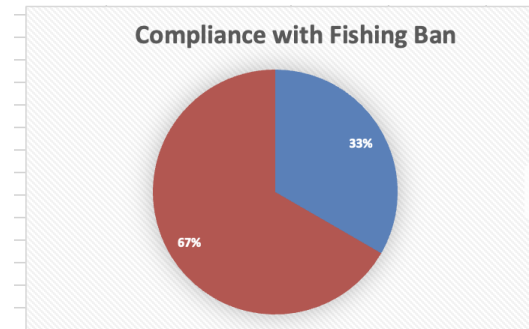
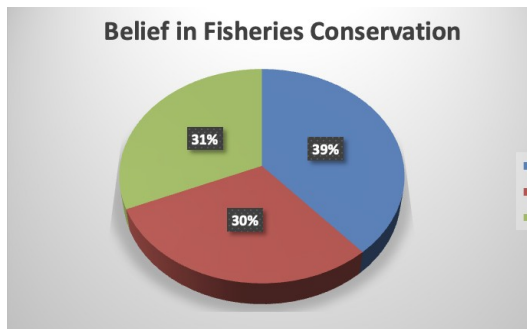
Catch and Fishing Productivity, Economic Dependence on Fisheries

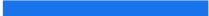


- Catch per trip varies among fishers, indicating differences in fishing capacity and effort.
- Fishers traveling farther offshore often report higher catches.
- Deeper fishing areas may have better fish availability and less competition.
- Most surveyed households rely heavily on fishing as their primary livelihood source.
- Income distribution shows strong economic dependence on fisheries.
- Income variation indicates financial instability and vulnerability within coastal fishing communities.

Awareness and Compliance with Regulations, Perceived Changes in Fish Stocks

- Many fishers are aware of fishing regulations, including seasonal bans and conservation measures.
- Compliance varies due to strong economic dependence on fishing livelihoods.
- Despite challenges, many respondents support conservation, indicating potential for community participation in sustainable fisheries management.
- Many fishers report declining fish availability and CPUE.
- Indicates increasing pressure on marine resources.
- Suggests possible overfishing and stock depletion.
- Highlights need for stronger fisheries management.
- Emphasizes importance of conservation and improved monitoring.





Implications and Conclusions

Fisheries are the primary livelihood for many coastal households in Balasore.

Fishers report declining fish availability and increasing economic vulnerability.

Awareness of sustainability regulations is mixed.

Strengthening community-based management and livelihood diversification is essential.

Future research should include broader regional studies and ecological data.

Reference

- Bennett, A., Coates, D., Gondwe, E., Shrestha, R. and Valbo-Jørgensen, J., 2025. *Livelihoods-based restoration of inland aquatic ecosystems for poverty reduction, food and nutrition security and biodiversity conservation: Guidance for policymakers and practitioners on putting livelihoods first in ecosystem restoration*. Food & Agriculture Org..
- Cooper, G., Dearing, J. and Willcock, S., 2025. Managing hysteretic recovery in lake systems.
- Das, A.R., Ema, S.J., Aktaruzzaman, M., Khanom, F. and Begum, K., 2025. Fishing Bans as a Conservation Tool: Socioeconomic Impacts and Livelihood Compensation in Practice. *Pathfinder of Research*, 3(2), pp.46-67.
- Das, P., Mandal, S., Nedungadi, P. and Raman, R., 2025. Unveiling sustainable tourism themes with machine learning based topic modeling. *Discover sustainability*, 6(1), p.280.
- Kamble, V.K., Brewer, D. and Oliver, M.E., 2025. Is There a Trade-Off between Forest Expansion and Agriculture?: Evidence from India. *Land Economics*, 101(3), pp.304-329.
- Kelkar, N., 2023. Inland capture fisheries, dam reservoirs, and protected areas for wildlife conservation in India: conflicts and ways forward. *Reviews in Fish Biology and Fisheries*, 33(1), pp.201-220.
- Kumar, P., Uthappa, A.R., Chavan, S.B., Chichaghare, A.R., Debta, H., Bhat, S. and Dagar, J.C., 2023. Achieving biodiversity conservation, livelihood security and sustainable development goals through agroforestry in coastal and island regions of India and Southeast Asia. In *Agroforestry for sustainable intensification of agriculture in Asia and Africa* (pp. 429-486). Singapore: Springer Nature Singapore.
- Roy, S.B. and Mukhopadhyay, R., 2022. Conceptual Framework for initiating forest mosaic integrated landscape management-IBRAD Model: Participatory Approach to ecosystem restoration for enhancement of ecosystem services.



Reference

- Sethi, T., Sethy, M. and Patra, S., 2025. Demographics to Livelihoods: A Study of Fishing Communities in Chilika Lake, Odisha.
- Singhar, A.S. and Gundimeda, H., 2024. Sustainable Harvests under different bio-economic scenarios of Chilika wetland's fisheries. *Ecology, Economy and Society-the INSEE Journal*, 7(1), pp.75-103.
- Tabares, A., Londoño-Pineda, A., Cano, J.A. and Gómez-Montoya, R., 2022. Rural entrepreneurship: An analysis of current and emerging issues from the sustainable livelihood framework. *Economies*, 10(6), p.142.
- Tessema, S.B. and Nayak, D., 2022. Analyzing the perceived prioritized forest ecosystem services under the participatory management: A case of Maksegnit District, Amhara Regional State, Ethiopia. *Trees, Forests and People*, 9, p.100318.
- Tripathy, P., 2025. Ecological-Economic Integration in Multi-species Fisheries: A Path toward Sustainable Marine Resource Management. *Asian Journal of Environment & Ecology*, 24(6), pp.176-186.
- Velmurugan, B., Samima, N., Banu, A.A., Safrin, J.H. and Balan, A., 2026. *From Catch to Consumer: Oceanic Wealth and Economic insights of Fisheries at State, National and Global Perspective*. Deep Science Publishing.

Thank You!

